Intelligent Agents

Lecture 2

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Intelligence: What Do We Want Out Of AI?

Key idea

Before we define an intelligent agent, we should ask what behavior we actually want from an AI system.

Image placeholder: ai-goals.jpg

1. Be independent from constant human control.
2. Solve problems rather than merely replay a fixed sequence.
3. Be as good as us—or better—on the task we care about.

In class

What do we mean by *better*? Faster? Safer? More accurate? More adaptable? More human-like?

Self-Driving Cars

- How does a self-driving car “see” the world?
- Is it following programmed rules, or adapting to new situations?
- How is this different from a simple remote-controlled car?

Image placeholder: `self-driving-car.jpg`

Key idea

The interesting part is not that the car moves. The interesting part is how it chooses what to do from what it perceives.

Vacuum Bot

- Your robot vacuum moves on its own, but is it thinking?
- How does it "see" the room?
- What kinds of choices does it make: straight-line movement, obstacle avoidance, room selection, charging?
- Is it following a preset routine, or does it adapt?

Image placeholder: `vacuum-bot.jpg`

In class

Which behaviors would convince you that the vacuum is doing more than replaying a script?

AI Agent: Comparing Intelligence

- A self-driving car navigates unpredictable streets.
- A vacuum bot maps and cleans a familiar space.
- Both perceive an environment and choose actions.
- Their environments differ dramatically in uncertainty, danger, scale, and consequence.

Image placeholder: `agent-comparison.jpg`

In class

Which one is more intelligent, and why? What evidence would justify the comparison?

Part 1: From Intelligence To Agents

Ada Lovelace's Argument (1843)

Example

"The Analytical Engine has no pretensions whatever to originate anything. It can do whatever we know how to order it to perform."

- A machine manipulates symbols according to rules, but does it think for itself?
- Is intelligence possible if every behavior ultimately comes from a mechanism?
- Does intelligence require originality, understanding, or self-direction?

In class

Are *you* an intelligent agent? What property makes the difference?

Image placeholder: `ada-lovelace.jpg`

Are Computers Intelligent?

Key idea

Even the physical substrate of computation is not fixed. Researchers have explored systems that combine biological neural tissue with electronic interfaces.

- The old lecture highlighted Cortical Labs' CL1 as a commercialized "biological computer."
- The system uses living human brain cells integrated with computing hardware and networked access.
- This raises a useful boundary question: if the hardware becomes more brain-like, does that make the system more intelligent?

In class

Does intelligence depend on what a system is made of, or on what it can do?

Image placeholder: `biological-computer.jpg`

Agent: A First Intuition

A philosophical intuition

An agent is something that appears to *think*: it perceives, reasons about what it perceives, and chooses actions in pursuit of goals.

- Perceives its environment.
- Reasons about its perceptions.
- Chooses actions to achieve goals.
- Connects naturally to the old philosophical idea *cogito, ergo sum*: "I think, therefore I am."

Key idea

In AI, we will make this idea operational: instead of proving that a system "really thinks," we study the relationship between percepts and actions.

Image placeholder: `agent-loop.jpg`

Agent: Sensors And Actuators

Definition

An **agent** perceives its environment through **sensors** and acts upon that environment through **actuators**.

Human Agent

Sensors: eyes, ears, touch, proprioception
Actuators: hands, legs, voice, facial movement

Robotic Agent

Sensors: cameras, lidar, infrared, encoders
Actuators: motors, wheels, grippers, speakers

Software Agent

Sensors: files, APIs, logs, network packets
Actuators: messages, database writes, API calls

Image placeholder: `agent-loop.jpg`

Percepts, Percept Sequences, And The Environment

Definition

A **percept** is what an agent senses at a particular moment. A **percept sequence** is the complete history of the agent's percepts.

- The current percept tells the agent what its sensors report *now*.
- The percept sequence preserves what has happened over time.
- The relevant **environment** is not necessarily the whole universe; it is the part of the world that matters to the agent's task.

Example

A vacuum robot's sequence might include: clean floor, dirt detected, wall ahead, turn left, low battery, charger detected.

Key idea

History matters whenever the same current percept can mean different things depending on what happened earlier.

Part 2: Agent Functions And Programs

Vacuum Bot: What Choices Does It Make?

Image placeholder: `vacuum-bot.jpg`

- Move forward, turn, or stop?
- Clean here or continue exploring?
- Avoid an obstacle or choose another route?
- Return to charge or keep working?
- Revisit an area or assume it is already clean?

In class

Which percepts should determine each choice?

Agent Function: The Abstract Model

Definition

The **agent function** maps every possible percept sequence to an action.

Image placeholder:

`agent-function-diagram.jpg`

- It is a theoretical, mathematical description of how an agent behaves.
- It specifies an action for every percept history the agent could encounter.
- Written literally as a table, it would be enormous and can be potentially infinite.

$$f : P^* \to A$$

Vacuum Cleaner Agent Function

Key idea

The abstract function must define an action for *every possible percept sequence*, not merely for the current sensor reading.

- Percepts might include room location, dirt status, obstacles, and battery state.
- The history might include every room visited and every earlier sensor reading.
- Possible actions include moving, turning, cleaning, stopping, and docking.

Example

The old vacuum-world table maps sequences such as (A, Dirty) or $(A, \text{Clean}), (B, \text{Dirty})$ to actions such as `Suck`, `Right`, or `Left`.

Image placeholder: `agent-function-table.jpg`

Agent Program: The Concrete Implementation

Definition

An **agent program** is the actual code or system that implements an agent function on a physical or software architecture.

- It runs on hardware and determines actions from percepts.
- It is *not* the same thing as the abstract agent function; it is one practical way to approximate or implement that function.
- A real program generalizes behavior rather than storing every possible percept sequence explicitly.

Example

Vacuum program: if dirt is detected, clean; else if an obstacle is detected, turn; otherwise move forward.

Image placeholder: `agent-program-visual.jpg`

Part 3: Rational Agents

Aristotle's Practical Reasoning

Example

"The function of man is an activity of the soul in accordance with reason."

Theoretical Reasoning

Knowing what is true.

Practical Reasoning

Deciding what to do in order to achieve a goal.

Image placeholder: `aristotle-practical.jpg`

Vacuum Bot: Is It Rational?

Image placeholder: `vacuum-rationality.jpg`

In class

Is a vacuum bot rational?

In class

Are *you* rational?

Key idea

Rationality is not about being human, conscious, or always correct. It is about choosing actions well relative to information and objectives.

Rationality Versus Omniscience

Omniscience — Impossible

A truly omniscient agent would know the *actual* outcome of every possible action before acting.

- It would know the future.
- It could always select the action with the best realized outcome.
- Real agents do not have this information.

Rationality — Possible

A rational agent chooses the action expected to maximize performance from the information available.

- Success is probabilistic, not guaranteed.
- A rational decision may still fail.
- What matters is the quality of the decision given the evidence.

Image placeholder: `rationality-omniscience-1.jpg`

Rationality Is Not Perfection

Rationality

Act on available information to maximize expected success.

Perfection

Always choose the action that turns out best—which would require advance knowledge of uncertain future events.

Key idea

AI does not need to be omniscient. It needs to be good enough at making decisions with limited knowledge.

Image placeholder:

`rationality-omniscience-2.jpg`

Part 4: Learning And Autonomy

Not Learning: The Danger Of Rigid Behavior

Key idea

Some agents do not need to learn if the world is perfectly predictable—but that makes them fragile.

Image placeholder: `rigid-robot.png`

- A fixed routine can be extremely effective in the environment it was designed for.
- A small change in the world can invalidate assumptions embedded in the routine.
- Without learning, the agent cannot improve from failure or adapt to new conditions.

Learning: Rational Agents Modify Future Percepts

- Actions should sometimes be taken not only to achieve an immediate goal, but also to gather useful information.
- Example: looking both ways before crossing the street improves the quality of the next decision.
- A rational agent must balance **acting** with **exploring**.

Image placeholder: `learning-loop.jpg`

Example

An action can change the world, but it can also change what the agent will be able to perceive and learn next.

Exploration In Unknown Environments

- A vacuum bot placed in an unknown room may need to explore before it can clean efficiently.
- It must learn where obstacles, walls, open regions, and charging locations are.
- Rational agents adapt from new percepts rather than relying entirely on pre-programmed knowledge.
- Exploration creates information that can improve future performance.

Image placeholder: `learning-exploration.jpg`

Key idea

Exploration is useful because knowledge itself can have future value.

Alan Turing On Learning (1950)

Example

"Instead of trying to produce a programme to simulate the adult mind, why not rather try to produce one which simulates the child's?"

- Turing suggested that intelligence need not be fixed at design time.
- Experience and environment can shape what a system becomes capable of doing.
- This aligns with the idea of agents that adapt rather than simply arriving "smart."

Image placeholder:

`turing-chappie.jpg`

Autonomy

Definition

Autonomy is the degree to which an agent relies on its own percepts and learning rather than only on pre-programmed knowledge.

- An agent that only follows hard-coded rules has limited autonomy.
- Experience can allow an agent to compensate for gaps in its initial knowledge.
- A fully rational agent should be able to adapt and learn when the world differs from its initial assumptions.

Image placeholder:

`autonomy-concept.jpg`

Autonomy: Prior Knowledge And Learning

Balancing prior knowledge and learning

Total autonomy from the start is not practical. An agent with no prior knowledge would have no reason to prefer one initial action over another.

The power of learning

With enough experience, an agent's behavior can become increasingly independent from its initial programming.

- Learning lets one intelligent system succeed across a wider variety of environments.
- Built-in knowledge gives the agent a useful starting point; experience improves it.

Image placeholder: `autonomy-comparison.jpg`

Part 5: Task Environments And PEAS

The Problem Agents Must Solve

Key idea

A **task environment** is the problem, and a rational agent is the proposed solution.

- Before designing an agent, we should define the environment in which it will operate.
- The **PEAS** framework forces us to specify the task clearly.

PEAS

Performance measure
Environment
Actuators
Sensors

Image placeholder: `peas-taxi.jpg`

PEAS Example: Self-Driving Taxi

Performance	Environment	Actuators	Sensors
Safety, passenger comfort, trip efficiency, obeying traffic laws, maximizing profits.	Roads, traffic, pedestrians, weather, passengers, obstacles.	Steering, acceleration, braking, signals, doors.	Cameras, lidar, GPS, speedometer, proximity sensors.

Key idea

PEAS is not an implementation. It is a disciplined specification of what the agent is supposed to accomplish and what information and actions are available.

PEAS Examples

Agent Type	Performance Measure	Environment	Actuators	Sensors
Water bottle fill station	Actually turns ON/OFF	Students, water bottles	Water valve	Proximity sensor
Grubhub delivery robot	On-time delivery, minimal spillage	Campus pathways, pedestrians, traffic	Wheels, delivery compartment door	GPS, cameras, lidar, proximity sensors
Code autograder	Accurate scoring, fair feedback	Student code submissions	Display of grades, feedback messages	Code input, test-case results
AC thermostat	Comfort, energy efficiency	A room or building	Heater/cooler switches, fan control	Temperature sensor, humidity sensor
Part-picking robot	Percentage of parts in correct bins	Conveyor belt with parts, bins	Jointed arm, gripper	Camera, tactile sensors, joint-angle sensors

Image placeholder: `peas-examples.jpg`

Part 6: Properties Of Task Environments

Observability: How Much Can The Agent See?

Fully Observable

The agent has complete information about the relevant environment state at all times.

Example

Chess: every piece is visible on the board.

Partially Observable

The agent receives limited, noisy, or incomplete information.

Example

Self-driving car: sensors cannot see around every corner.

Unobservable

The agent has no direct perceptual access to the relevant environment state.

Example

A robot attempting to navigate with no usable sensing.

Image placeholder: `observability.jpg`

Single-Agent Versus Multi-Agent

Single-Agent

The agent operates alone; other decision-making agents do not significantly affect the task.

Example

Vacuum cleaner in an empty house.

Multi-Agent

Other agents exist and their choices influence the environment.

Example

A swarm of delivery drones, traffic, games, or markets.

Competitive

Agents work against one another, as in chess or trading.

Cooperative

Agents coordinate toward shared objectives, as in robot teams.

Deterministic Versus Stochastic

Deterministic

The next state is fully determined by the current state and the agent's action.

Example

Sliding puzzle: under the model, moving a tile has a definite result.

Stochastic / Nondeterministic

Outcomes are uncertain; the same state and action can lead to different results.

Example

Poker: hidden cards and uncertain opponent behavior affect outcomes.

Image placeholder: `deterministic-stochastic.jpg`

Episodic Versus Sequential

Episodic

Each decision is independent; earlier actions do not affect later decisions.

Example

Image classification: each image can be analyzed separately.

Sequential

Future decisions depend on previous actions and the states they created.

Example

Video tracking, driving, games, and navigation.

Image placeholder: `episodic-sequential.jpg`

Static Versus Dynamic

Static

The environment does not change while the agent deliberates unless an agent acts.

Example

Chess without a clock: the board does not change on its own.

Dynamic

The world can change while the agent is deciding.

Example

Self-driving car: traffic continues moving while the car plans.

Image placeholder: `static-dynamic.jpg`

Discrete Versus Continuous

Discrete

The relevant states, percepts, actions, or time steps form a finite or countable set.

Continuous

States or actions vary over continuous ranges.

Example

Chess: legal moves are discrete alternatives.

Example

A robotic arm can vary joint positions and velocities continuously.

Image placeholder: `discrete-continuous.jpg`

Known Versus Unknown

Known

The agent understands the rules that determine how actions affect the environment.

Example

Chess AI: legal moves and game rules are known.

Unknown

The agent must learn how the world works through experience.

Example

A newly built robot exploring unfamiliar terrain while localizing itself.

Image placeholder: `known-unknown.jpg`

Environments?

Task Environment	Observable	Agents	Deterministic	Episodic	Static	Discrete
Crossword puzzle	Fully	Single	Deterministic	Sequential	Static	Discrete
Chess with a clock	Fully	Multi	Deterministic	Sequential	Semi	Discrete
Poker	Partially	Multi	Stochastic	Sequential	Static	Discrete
Backgammon	Fully	Multi	Stochastic	Sequential	Static	Discrete
Taxi driving	Partially	Multi	Stochastic	Sequential	Dynamic	Continuous
Medical diagnosis	Partially	Single	Stochastic	Sequential	Dynamic	Continuous
Image analysis	Fully	Single	Deterministic	Episodic	Semi	Continuous
Part-picking robot	Partially	Single	Stochastic	Episodic	Dynamic	Continuous
Refinery controller	Partially	Single	Stochastic	Sequential	Dynamic	Continuous
English tutor	Partially	Multi	Stochastic	Sequential	Dynamic	Discrete

Key idea

These labels are modeling assumptions. A task can move between categories when we change what counts as the state, what sensors are available, or what uncertainty we include.

Part 7: Course Reminder

Reminder

Lab 0

Break yourselves into groups.

Final Project

Milestone 1: Team Formation + Idea